RA2311047010044

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NAME K.D.D.Sai Abhiram SUBJECT Deep learning lab obs

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1		Analyize and exploring the deep Leanring Platforms	24-07-25	[illegible]
2		Implement a classifier using open-Source data set	7/8/25	[illegible]
3.		study of the classifiers with respect to Statistical parameters	7/8/25	[illegible]
4.		Build a simple feed forward neural network to recognize hand written characters	14/8/25	[illegible]
5.		Study of activation function and its role	28-8-25	[illegible]
6.		Implement gradient descent & back Propagation in deep neural network.	09-9-25	[illegible]
7.		Build a CNN model to classify Cats & dogs Images	16-9-25	[illegible]
8.		Experiment using LSTM		[illegible]
9.		Build a RNN		
10		Perform Compression on MNIST data set using Autoencoder		[illegible]
11.		to implement experiment using variational Autoencoder		
12.		Implement DCGAN to generate Colour images	27/10	[illegible]
13.		understanding the architecture of Pre-trained model	27/10	
14.		Pre-trained CNN model as feature extractor using TLM		
15.		Implement a Yolo model to detect objects		

completed [illegible]

15. Implement a yolo model to detect objects

Aim: To implement the Yolo model for real time object detection in images

Objectives:

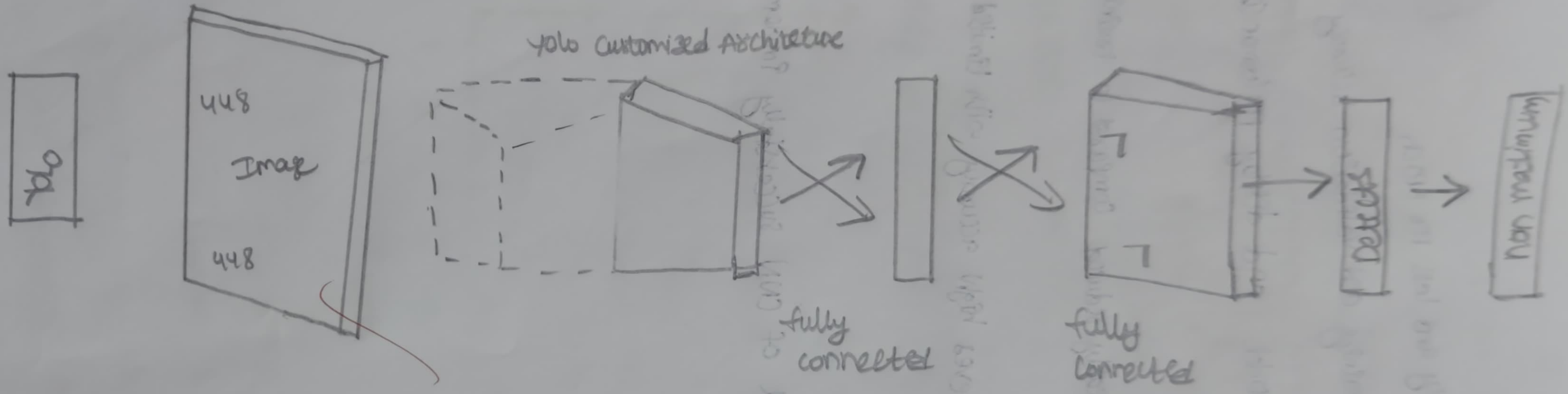
1. To understand the working of yolo architecture for object detection
2. To detect & localize multiple objects in an image
3. to apply pre-trained yolo weights on a sample dataset (or) image
4. to visualize bounding boxes & confidence scores

~~Algorithm~~

Pseudocode:

```
Load yolo pre-trained model
Load class labels
for each image:
    Preprocess (resize, normalize)
    Pass through yolo model
    Get bounding boxes
    Draw boxes & labels
    display detect images
```

Yolo Architecture model



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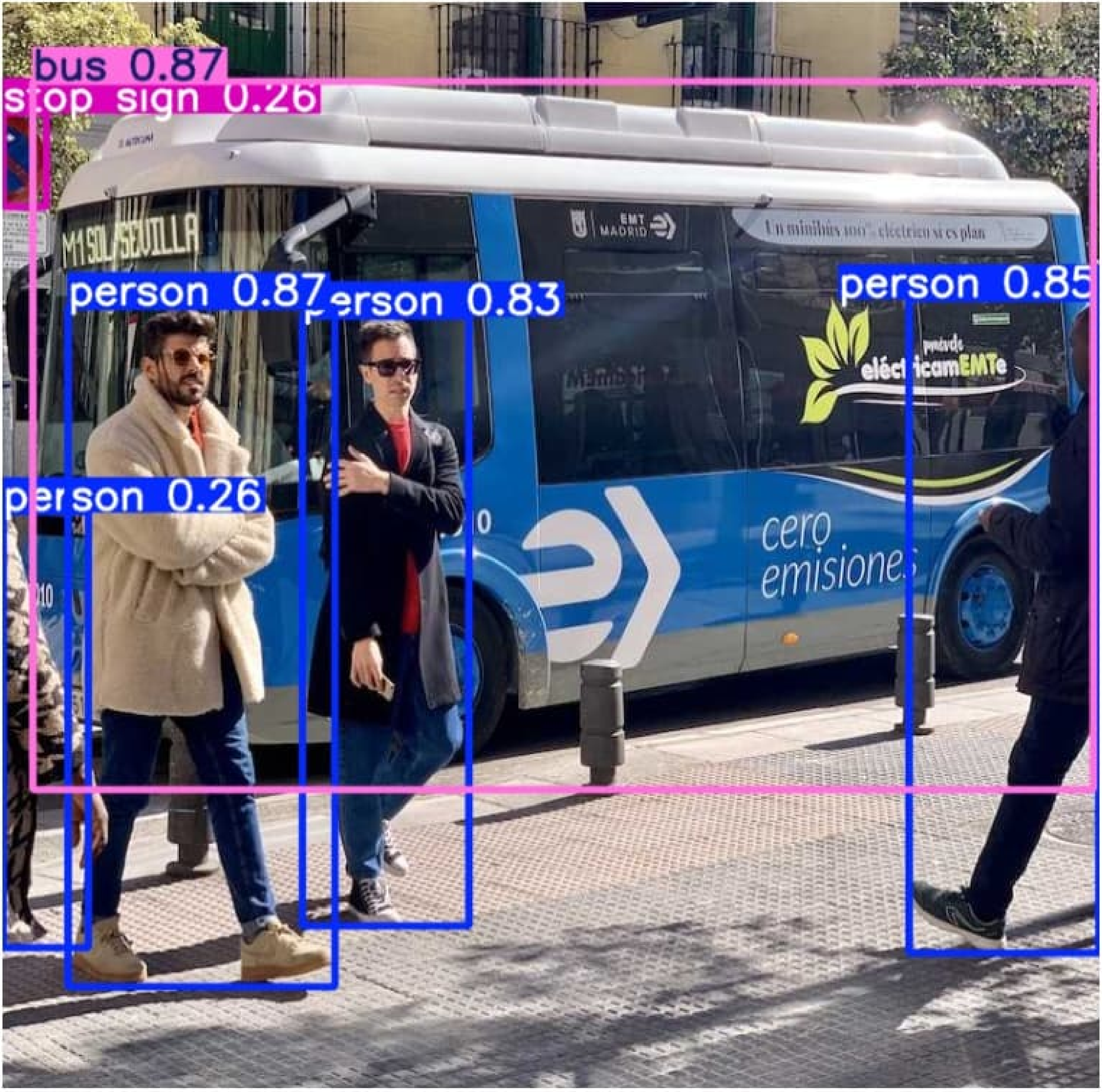
```
# Algorithm
# Install YOLOv8
# Load pretrained model
# Perform detection on sample image
# View results
#####
!pip install ultralytics
from ultralytics import YOLO
model = YOLO('yolov8n.pt')  # nano model
results = model('https://ultralytics.com/images/bus.jpg')

results[0].show()
```

```
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralytics) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.8.0->ultralytics) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.8.0->ultralytics) (3.0.3)

Downloading https://ultralytics.com/images/bus.jpg to 'bus.jpg': 100% ━━━━━━━━━━ 134.2KB 6.2MB/s 0.0s
image 1/1 /content/bus.jpg: 640x480 4 persons, 1 bus, 1 stop sign, 393.3ms
Speed: 30.8ms preprocess, 393.3ms inference, 37.3ms postprocess per image at shape (1, 3, 640, 480)
```



bus 0.87
stop sign 0.26
M1 SOL/SEVILLA
EMT MADRID
person 0.87
person 0.83
person 0.85
eléctricamEMTe
person 0.26
cero emisiones

Output

448X640 (no detections), 68.1ms

Speed: 10.7 ms preprocess, 68.1 ms inference

0.8ms post process per image at shape (1, 3, 448 640)

Observation

* Yolo detects multiple objects with boundary boxes and confidence scores

* Inference is fast and efficient, suitable for real time application

* model accurately detects objects

*

Result:

the Yolo model successfully implemented to detect the objects.